Grade 9 Mathematics – Algebraic Fractions

Answer Sheet – Worked Solutions

Chalaka Perera | BSc (Hons) Computer Engineering, University of Peradeniya

Remember: To add or subtract algebraic fractions, first make the denominators equal using the LCM (least common multiple) of the denominators, then add or subtract the numerators. Put brackets around numerators with more than one term before subtracting, and always simplify the final answer.

Question 1

(i) $\frac{3x}{5}$ (ii) $\frac{3a}{9} = \mathbf{\frac{a}{3}}$ (iii) $\frac{x + 2 + 2x - 3}{7} = \mathbf{\frac{3x - 1}{7}}$ (iv) $\frac{3x + 1 - (x - 5)}{4} = \frac{2x + 6}{4} = \mathbf{\frac{x + 3}{2}}$

Question 2

(i) $\frac{3x + 2x}{6} = \mathbf{\frac{5x}{6}}$ (ii) $\frac{9a - 2a}{12} = \mathbf{\frac{7a}{12}}$ (iii) $\frac{4(x + 1) + 3(x - 2)}{12} = \frac{4x + 4 + 3x - 6}{12} = \mathbf{\frac{7x - 2}{12}}$

(iv) $\frac{2(2x - 1) - 5(x - 3)}{10} = \frac{4x - 2 - 5x + 15}{10} = \mathbf{\frac{13 - x}{10}}$ (v) $\frac{6x + 4x + 3x}{12} = \mathbf{\frac{13x}{12}}$

Question 3

(i) $\mathbf{\frac{8}{x}}$ (ii) $\frac{4}{2a} = \mathbf{\frac{2}{a}}$ (iii) $\frac{x + 1 + x - 1}{y} = \mathbf{\frac{2x}{y}}$

Question 4

(i) $\frac{4}{2x} + \frac{3}{2x} = \mathbf{\frac{7}{2x}}$ (ii) $\frac{b}{ab} + \frac{a}{ab} = \mathbf{\frac{a + b}{ab}}$ (iii) $\frac{10}{6x} - \frac{1}{6x} = \frac{9}{6x} = \mathbf{\frac{3}{2x}}$

(iv) $\frac{9}{12p} + \frac{8}{12p} = \mathbf{\frac{17}{12p}}$ (v) $\frac{2x}{x^2} - \frac{3}{x^2} = \mathbf{\frac{2x - 3}{x^2}}$ (vi) $\frac{3a^2}{6ab} + \frac{2b^2}{6ab} = \mathbf{\frac{3a^2 + 2b^2}{6ab}}$

Question 5

(i) $\frac{x}{2x} + \frac{2}{2x} = \mathbf{\frac{x + 2}{2x}}$ (ii) $\frac{x^2}{3x} - \frac{6}{3x} = \mathbf{\frac{x^2 - 6}{3x}}$ (iii) $\frac{3x}{x} + \frac{2}{x} = \mathbf{\frac{3x + 2}{x}}$

Question 6

(i) The student added the numerators and also added the denominators. Fractions can only be added after making the denominators equal (LCM of 2 and 3 is 6).

(ii) $\frac{3x}{6} + \frac{2x}{6} = \mathbf{\frac{5x}{6}}$

Question 7

(i) $\frac{3(x+3)-2(x-1)}{6} = \frac{3x+9-2x+2}{6} = \mathbf{\frac{x+11}{6}}$

(ii) $\frac{1+11}{6} = \frac{12}{6} = \mathbf{2}$

(iii) $\frac{1+3}{2} - \frac{1-1}{3} = 2 - 0 = 2$ (the answers agree)

Question 8

(i) $\frac{x}{3} + \frac{x}{4} = \frac{4x+3x}{12} = \mathbf{Rs\ \frac{7x}{12}}$

(ii) $x - \frac{7x}{12} = \frac{12x-7x}{12} = \mathbf{Rs\ \frac{5x}{12}}$

(iii) $\frac{5 \times 1200}{12} = \mathbf{Rs\ 500}$

Question 9

(i) Perimeter $= 2(\frac{x+1}{2} + \frac{x-1}{3}) = 2 \times \frac{3(x+1)+2(x-1)}{6} = 2 \times \frac{5x+1}{6} = \mathbf{\frac{5x+1}{3}\ cm}$

(ii) $\frac{5 \times 7 + 1}{3} = \frac{36}{3} = \mathbf{12\ cm}$. Check: length = 4 cm, breadth = 2 cm, perimeter = 2(4 + 2) = 12 cm ✔

Question 10

(i) $\frac{2(x+2)-(x-4)}{6} = \frac{2x+4-x+4}{6} = \mathbf{\frac{x+8}{6}}$

(ii) $\frac{x+8}{6} = 3 \rightarrow x + 8 = 18 \rightarrow \mathbf{x = 10}$

Question 11

(i) $\frac{a}{4} = \frac{3a}{12}$, so the second fraction must be $\frac{4b}{12} = \frac{2b}{6}$ → missing term = **2b**

(ii) $\frac{\square}{x} = \frac{5}{3x} + \frac{1}{3x} = \frac{6}{3x} = \frac{2}{x}$ → missing term = **2**